

Arduino – Light Sensor (LDR) Example

Example: Turn ON an LED when it gets dark

The LDR is connected to analog pin A0. Arduino reads the light level using `analogRead()`. If the reading is below 300, the LED turns ON.

```
int lightSensor = A0;
int led = 13;

void setup()
{
    pinMode(led, OUTPUT);
}

void loop()
{
    int light = analogRead(lightSensor);

    if (light < 300)
    {
        digitalWrite(led, HIGH);
    }
    else
    {
        digitalWrite(led, LOW);
    }
}
```

How it works:

1. Arduino reads the LDR value from A0.
2. If the value is less than 300, the LED turns ON.
3. If the value is 300 or higher, the LED turns OFF.

System flow: LDR → A0 → ADC → Arduino → LED